

SIXIANG (SEAN) CHEN

Ph.D. Student @ HKUST(GZ) ROAS · Generative Models / Reward Modeling / Vision Agents

ephemeral182@gmail.com | [ephemeral182.github.io](https://github.com/ephemeral182) | [Google Scholar \(1800+ Citations\)](https://scholar.google.com/citations?user=ephemeral182) | +86-182-5916-6302

RESEARCH SUMMARY

Second-year Ph.D. student at HKUST(GZ) ROAS, advised by Prof. Lei Zhu and Prof. Fugee Tsung. My research lies at the intersection of **Generative Models**, **Reward Modeling**, **Vision Agents**, and **Visual Perception**. I build generalized assistants for aesthetic poster creation, design reward models for human-aligned generation, and develop agentic systems for open-world reasoning.

- **Aesthetic Poster Assistants:** Lead the development of **PosterCraft [ICLR 2026]**, which is widely adopted by the community, and **PosterOmni [CVPR 2026]**, which has been deployed for internal industrial applications.
- **Reward Modeling for Design:** Develop **PosterReward [CVPR 2026]** to unlock accurate evaluation signals, significantly boosting ordinary open-source text-to-image models to top-tier aesthetic performance, serving as a core reward model for internal training.
- **Agents & Reasoning:** Explore multimodal agents for adverse-scene understanding and restoration (e.g., **JarvisIR [CVPR 2025]** and **RestoreAgent [NeurIPS 2024]**), focusing on unified reasoning capabilities.
- **Diffusion Priors for Perception:** Leverage diffusion priors for restoration and perception tasks, including **GenHaze [ICCV'25]**, **AGLLDiff [AAAI 2025]**, **DiffWeather [ECCV 2024]** and **DTPM [CVPR 2024 Highlight]**.

EDUCATION

The Hong Kong University of Science and Technology (Guangzhou)

Ph.D. Student, Robotics and Autonomous Systems (ROAS)

Guangzhou, China

Sep. 2024 – Present

- Advisors: Prof. Lei Zhu; Prof. Fugee Tsung.
- Research: Generative Models, Reward Modeling, Vision Agents, Image Restoration.

Jimei University

B.Eng. in Communication / Telecommunication Engineering

Xiamen, China

Sep. 2019 – Jun. 2023

- Selected coursework: Complex Variable (100/100), C Programming (98/100), Mathematical Modeling (96/100), Probability & Statistics (96/100).
- Advisor: Prof. Erkang Chen. **Awards:** Outstanding Graduate.

RESEARCH EXPERIENCE

The Hong Kong University of Science and Technology (Guangzhou)

Research Assistant, ROAS

Ph.D. Student, ROAS

Guangzhou, China

Jul. 2023 – Aug. 2024

Sep. 2024 – Present

- **Diffusion Priors for Perception:** Leverage diffusion priors for robust restoration and perception tasks, publishing works such as **GenHaze [ICCV'25]**, **AGLLDiff [AAAI 2025]**, **DiffWeather [ECCV 2024]** and **DTPM [CVPR 2024 Highlight]**.
- **Agents & Unified Reasoning:** Develop multimodal agentic systems for adverse-scene understanding and restoration, including **JarvisIR [CVPR 2025]** and **RestoreAgent [NeurIPS 2024]**, focusing on unified reasoning capabilities.

Meigen-AI, Meituan

Research Scientist Intern (Beidou Talent Program)

Shanghai/Shenzhen, China

Apr. 2025 – Present

- **Aesthetic Poster Generation:** Lead the development of **PosterCraft [ICLR 2026]** (widely adopted by the community) and **PosterOmni [CVPR 2026]** (deployed for internal industrial applications), unifying local edits and global design for high-quality creation.
- **Reward Modeling:** Develop **PosterReward [CVPR 2026]** to unlock accurate evaluation signals for aesthetics and typography, significantly boosting ordinary open-source models to top-tier performance, serving as a core reward model for internal training.

SELECTED PUBLICATIONS

Summary: Published **20+** papers in top-tier conferences/journals (CVPR, ICCV, ECCV, ICLR, NeurIPS, AAAI, ACM MM), including **11** first-author/co-first-author papers.

* *Equal Contribution.* ✉ *Corresponding Author*

First-Author & Co-First-Author Publications

[CVPR 2026] PosterOmni: Generalized Artistic Poster Creation via Task Distillation and Unified Reward Feedback
Sixiang Chen*, Jianyu Lai*, Jialin Gao*, Hengyu Shi*, Zhongying Liu*, Tian Ye, Junfeng Luo, Xiaoming Wei, Lei Zhu.

[CVPR 2026] PosterReward: Unlocking Accurate Evaluation for High-Quality Graphic Design Generation
Jianyu Lai*, Sixiang Chen*, Jialin Gao*, Hengyu Shi, Zhongying Liu, Fuxiang Zhai, Junfeng Luo, Xiaoming Wei, Lujia Wang, Lei Zhu.

[ICLR 2026] PosterCraft: Rethinking High-Quality Aesthetic Poster Generation in a Unified Framework
Sixiang Chen*, Jianyu Lai*, Jialin Gao*, Tian Ye, Haoyu Chen, Hengyu Shi, Shitong Shao, Yunlong Lin, Song Fei, Zhaohu Xing, Yeying Jin, Junfeng Luo, Xiaoming Wei, Lei Zhu.

[Tech Report 2025] An Empirical Study of GPT-4o Image Generation Capabilities
Sixiang Chen*, Jinbin Bai*, Zhuoran Zhao*, Tian Ye*, Qingyu Shi, Donghao Zhou, Wenhao Chai, Xin Lin, Jianzong Wu, Chao Tang, Shilin Xu, Tao Zhang, Haobo Yuan, Yikang Zhou, Wei Chow, Linfeng Li, Xiangtai Li, Lei Zhu, Lu Qi.

[ICCV 2025] GenHaze: One-step Controllable Haze Generation for Real-World Dehazing
Sixiang Chen, Tian Ye, Yunlong Lin, Yeying Jin, Yijun Yang, Haoyu Chen, Jianyu Lai, Song Fei, Zhaohu Xing, Fugee Tsung, Lei Zhu.

[CVPR 2025] SnowMaster: Comprehensive Real-world Image Desnowing via MLLM with Multi-Model Feedback Optimization
Jianyu Lai*, Sixiang Chen*, Yunlong Lin, Tian Ye, Yun Liu, Song Fei, Zhaohu Xing, Hongtao Wu, Wei Wang, Lei Zhu.

[AAAI 2025] AGLLDiff: Guiding Diffusion Models Towards Unsupervised Training-Free Real-World Low-Light Image Enhancement
Yunlong Lin*, Tian Ye*, Sixiang Chen*, Zhenqi Fu, Yingying Wang, Wenhao Chai, Zhaohu Xing, Lei Zhu, Xinghao Ding.

[ECCV 2024] Teaching Tailored to Talent: Adverse Weather Restoration via Prompt Pool and Depth-Anything Constraint
Sixiang Chen, Tian Ye, Kai Zhang, Zhaohu Xing, Yunlong Lin, Lei Zhu.

[ICCV 2023] Adverse Weather Removal with Codebook Priors
Tian Ye*, Sixiang Chen*, Jinbin Bai*, Shi Jun, Chenghao Xue, Jingjia Jiang, Junjie Yin, Erkang Chen, Yun Liu.

[ICCV 2023] Sparse Sampling Transformer with Uncertainty-Driven Ranking for Unified Removal of Raindrops and Rain Streaks
Sixiang Chen*, Tian Ye*, Jinbin Bai, Jun Shi, Erkang Chen, Lei Zhu.

[ACM MM 2023] Uncertainty-Driven Dynamic Degradation Perceiving and Background Modeling for Efficient Single Image Desnowing
Sixiang Chen, Tian Ye, Chenghao Xue, Haoyu Chen, Yun Liu, Erkang Chen, Lei Zhu.

[ACM MM 2023] CPLFormer: Cross-scale Prototype Learning Transformer for Image Snow Removal
Sixiang Chen, Tian Ye, Yun Liu, Jinbin Bai, Haoyu Chen, Yunlong Lin, Jun Shi, Erkang Chen.

Co-Authored Publications

[CVPR 2025] JarvisIR: Elevating Autonomous Driving Perception with Intelligent Image Restoration
Yunlong Lin*, Zixu Lin*, Haoyu Chen*, Panwang Pan*, Chenxin Li, Sixiang Chen, Kairun Wen, Yeying Jin, Wenbo Li, Xinghao Ding.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

[CVPR 2025] Detect Any Mirrors: Boosting Learning Reliability with an Iterative Data Engine
Zhaohu Xing, Lihao Liu, Yijun Yang, Hongqiu Wang, Tian Ye, Sixiang Chen, Wenxue Li, Guang Liu, Lei Zhu.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

[AAAI 2025] PromptHaze: Prompting Real-world Dehazing via Depth Anything
Tian Ye, Sixiang Chen, Haoyu Chen, Wenhao Chai, Jingjing Ren, Zhaohu Xing, Wenxue Li, Lei Zhu.
AAAI Conference on Artificial Intelligence (AAAI), 2025.

[AAAI 2025] Unsupervised Low-light Image Enhancement with Lookup Tables and Diffusion Priors
Yunlong Lin, Zhenqi Fu, Kairun Wen, Tian Ye, Sixiang Chen, Ge Meng, Yingying Wang, Yue Huang, Xiaotong Tu, Xinghao Ding.
AAAI Conference on Artificial Intelligence (AAAI), 2025.

[NeurIPS 2024] RestoreAgent: Autonomous Image Restoration Agent via Multimodal Large Language Models
Haoyu Chen, Wenxue Li, Jinjin Gu, Jingjing Ren, Sixiang Chen, Tian Ye, Renjing Pei, Kaiwen Zhou, Fenglong Song, Lei Zhu.

Conference on Neural Information Processing Systems (NeurIPS), 2024.

[CVPR 2024 Highlight] Learning Diffusion Texture Priors for Image Restoration

Tian Ye, Sixiang Chen, Wenhao Chai, Zhaoxu Xing, Jing Qin, Ge Lin, Lei Zhu.

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

[ACM MM 2023] Nighthazeformer: Single nighttime haze removal using prior query transformer

Yun Liu, Zhongsheng Yan, Sixiang Chen[✉], Tian Ye, Wenqi Ren, Erkang Chen.

ACM International Conference on Multimedia (ACM MM), 2023.

[ECCV 2022 Oral] Perceiving and Modeling Density for Image Dehazing

Tian Ye*, Yunchen Zhang*, Mingchao Jiang*, Liang Chen, Yun Liu, Sixiang Chen, Erkang Chen.

European Conference on Computer Vision (ECCV), 2022.

HONORS & ACADEMIC SERVICE

Honors & Awards

- **Third Place**, NTIRE 2025 Challenge on Day and Night Raindrop Removal (CVPR Workshop)
- **Fifth Place**, NTIRE 2024 Challenge on Low Light Image Enhancement (CVPR Workshop)
- Postgraduate Scholarship of HKUST(GZ), 2024
- Outstanding Graduate, Jimei University, 2023
- International Third Prize (Team Leader), Mathematical Contest In Modeling, 2022
- National Second Prize (Team Leader), China Undergraduate Mathematical Contest in Modeling, 2021
- National Second Prize (Team Leader), Mathorcup Mathematical Contest in Modeling, 2021

Academic Service

- **Conference Reviewer:** CVPR (2025, 2026), ICCV (2023, 2025), ECCV (2024), NeurIPS (2024, 2025), ICLR (2025, 2026), ICML (2025), AAAI (2025), ACM MM (2023, 2024), ICRA (2023), ACCV (2022).
- **Journal Reviewer:** IEEE TPAMI, IEEE TIP, IEEE TMM, IEEE TCSVT, IEEE TITS.